



3305 - Analyzing Giant Clumps in JWST Images of Star-Forming Galaxies to Constrain Feedback

Cycle: 2, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

A majority of star-forming galaxies at redshifts $z > 1$ have giant stellar clumps that are much more massive than star clusters in nearby galaxies. The masses and lifetimes of such clumps in high-resolution galaxy simulations strongly depend on the supernova feedback assumed, which remains one of the main uncertainties in galaxy formation theory. Accurate measurements of the observed clump stellar masses and lifetimes can provide new and independent constraints on feedback models. NIRCам data will dramatically improve on HST in both resolution and depth, as well as opening the window of rest-frame optical and even IR emission beyond $z \sim 3$. This improvement will allow measurement with unprecedented accuracy of the clump stellar population properties in $z \sim 1-3$ galaxies, and at even higher redshifts up to $z=7$. We propose to use JWST/CEERS survey to detect and analyze clumps by using two methods: (1) standard source detection and (2) novel machine learning (ML), both together providing a comprehensive view of clumps. In order to train ML, we will generate training sets of mock images at JWST resolution and filters. Besides the accurate measurements of clumps and strong constraints on feedback models, the proposed research will also provide the following value-added products: (1) a public catalog of stellar mass, age, dust extinction, star formation rate, and their uncertainties of clumps in galaxies at $z \sim 1-7$, derived with Bayesian SED fitting; (2) a public database of mock NIRCам images of state-of-the-art cosmological simulations with corresponding simulation metadata; and (3) all codes and documentation created for the analysis of JWST images via GitHub.

OBSERVING DESCRIPTION

This is an archival proposal. We propose to use published CEERS NIRCам images (F115W, F200W, F277W, F356W, and F444W) to detect giant star-forming clumps at $z=1\sim7$. We will use both the standard source detection method and novel machine learning to detect and measure clump properties. We will use SED-fitting to derive physical parameters of clumps and compare them to the state-of-the-art simulations. The project will also generate and publish mock NIRCам images from cosmological simulations.